SPA617
RADIO ASTRONOMY LABORATORY

Experiment 2

RF Components: Characterization and Measurement

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1 Aim

To understand the operation and experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in an RF receiver chain.

2 Objectives

1. To characterize an RF filter, identify its type, and determine its cutoff frequency and stop-band frequency.
2. To demonstrate a quarter-wave open-circuit stub as a notch filter.
3. To observe frequency conversion in an RF mixer and estimate its conversion loss.
4. To measure the power division and insertion loss of a power splitter/combiner.
5. To measure amplifier gain, the 1-dB compression point, and the third-order intercept point.
6. To estimate the input reflection coefficient and return loss of the filter.

3 Theory

3.1 RF Components in a Receiver Chain

A basic RF receiver chain consists of several components. A filter selects or rejects specified frequency ranges. A mixer translates a signal from one frequency to another. A low-noise amplifier amplifies weak received signals while adding as little noise as possible. A power splitter or combiner distributes a signal between multiple paths or combines signals from different paths.

3.2 Filter Transmission Characteristics

A filter passes selected frequency components and attenuates others. The commonly used filter types are:

- Low-pass filter: passes frequencies below the cutoff frequency.
- High-pass filter: passes frequencies above the cutoff frequency.
- Band-pass filter: passes frequencies between two cutoff frequencies.
- Band-reject or notch filter: attenuates frequencies within a specified stopband.

The insertion loss is

$$IL = P_{\text{in}} - P_{\text{out}}. \tag{1}$$

The cutoff frequency f_c is the frequency at which the insertion loss has increased by 3 dB relative to the passband level.

3.3 Reflection Coefficient and Return Loss

When the input impedance of the filter is not equal to the characteristic impedance of the RF system, part of the incident signal is reflected. The reflection coefficient is

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}. \tag{2}$$

The return loss is

$$RL = -20 \log_{10} |\Gamma|. \tag{3}$$

In the Tee-connector measurement,

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2}, \qquad V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2}, \tag{4}$$

so that

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}, \qquad \text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{5}$$

3.4 Quarter-Wave Open-Circuit Stub

An open-ended coaxial stub connected in shunt behaves as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end is transformed approximately into a low impedance at the Tee junction, producing a notch. The expected notch frequency is

$$f_0 = \frac{VF\, c}{4L}, \tag{6}$$

where VF is the velocity factor, c is the speed of light and L is the physical length of the stub.

3.5 Mixer

An RF mixer is a three-port nonlinear device consisting of RF, local oscillator (LO), and intermediate frequency (IF) ports. The mixer produces sum and difference frequencies:

$$f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}}, \qquad f_{\text{IF}} = |f_{\text{RF}} - f_{\text{LO}}|. \tag{7}$$

The conversion loss is

$$L_{\text{conv}} = P_{\text{RF}} - P_{\text{IF}}. \tag{8}$$

3.6 Power Splitter/Combiner

An ideal two-way equal power splitter divides the input power equally between two output ports, so each output is approximately 3 dB below the input:

$$L_1 = P_{\text{in}} - P_{\text{out1}}, \qquad L_2 = P_{\text{in}} - P_{\text{out2}}. \tag{9}$$

Used in reverse as a combiner with two coherent in-phase signals of the same frequency, the voltages add, and with V the peak voltage across $R = 50\ \Omega$,

$$P = \frac{V^2}{2R}. \tag{10}$$

3.7 RF Amplifier

The amplifier gain is

$$G = P_{\text{out}} - P_{\text{in}}. \tag{11}$$

At sufficiently high input power, the gain decreases. The 1-dB compression point is the input power at which the gain has fallen by 1 dB from its small-signal value.

3.8 Third-Order Intercept Point

For two input frequencies f_1 and f_2, the third-order intermodulation products closest to the fundamentals are $2f_1 - f_2$ and $2f_2 - f_1$. The fundamental output follows approximately a 1:1 slope, whereas the third-order product follows approximately a 3:1 slope. The extrapolated intersection gives the third-order intercept point (TOI). With fixed slopes, $P_{\text{fund}} = P_{\text{in}} + C_1$ and $P_{\text{IM3}} = 3P_{\text{in}} + C_3$, so

$$P_{\text{in,TOI}} = \frac{C_3 - C_1}{1 - 3}, \qquad P_{\text{out,TOI}} = P_{\text{in,TOI}} + C_1. \tag{12}$$

4 Apparatus

- RF signal generator(s)
- Spectrum analyser
- Digital storage oscilloscope
- RF filter
- RF mixer
- Power splitter/combiner
- RF amplifier
- RF Tee connector
- Open-ended coaxial stub
- 50 Ω terminations
- RF cables, adapters and attenuators

5 Experimental Procedure

5.1 Filter Characterization

A short reference cable was first connected between the signal generator and spectrum analyser. The input power was measured at the selected frequencies. The filter was then inserted without changing the generator setting, cables or adapters. The output power was measured at the same frequencies and the insertion loss was calculated using $IL = P_{\text{in}} - P_{\text{out}}$.

5.2 Filter Return Loss

The filter was connected using a Tee connector. The oscilloscope input impedance was set to 1 MΩ. The maximum and minimum voltage amplitudes at the Tee junction were recorded while sweeping the frequency. The incident voltage, reflected voltage, reflection coefficient and return loss were then calculated.

5.3 Quarter-Wave Stub

An open-ended coaxial stub was connected to the Tee connector. The expected notch frequency was calculated from $f_0 = VF\,c/4L$. The power response was measured without and with the stub. The frequency corresponding to the maximum attenuation was identified as the measured notch frequency.

5.4 Mixer

RF and LO signals were applied to the mixer and the output was observed using the spectrum analyser. The sum and difference frequencies were calculated and compared with the measured IF frequency. The conversion loss was calculated.

5.5 Power Splitter and Combiner

The splitter input was connected to the signal generator. Each output was measured separately using the spectrum analyser while the unused output was terminated with a 50 Ω load. For the combiner, two signals of equal frequency were applied to the two input ports and the output power was measured and compared with the ideal coherent sum.

5.6 Amplifier

The amplifier was powered using the specified DC supply. The input and output powers were measured over the operating frequency range, and the gain was calculated using $G = P_{\text{out}} - P_{\text{in}}$. The input power was subsequently increased at a fixed frequency to observe gain compression and determine the 1-dB compression point. Finally, two closely spaced input frequencies ($f_1 = 10$ MHz, $f_2 = 13$ MHz) were applied to measure the third-order intermodulation products at $2f_1 - f_2 = 7$ MHz and $2f_2 - f_1 = 16$ MHz and estimate the TOI.

6 Results and Analysis

6.1 Filter Transmission

Table 1: Filter transmission measurements.

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	IL (dB)
100	−0.90	−1.36	0.46
150	−0.90	−1.59	0.69
200	−0.90	−1.84	0.94
250	−0.90	−1.70	0.80
300	−0.90	−1.89	0.99
350	−0.90	−2.50	1.60
400	−0.90	−2.22	1.32
450	−0.90	−3.09	2.19
500	−0.90	−3.26	2.36
550	−0.90	−4.72	3.82
600	−0.90	−8.79	7.89
650	−0.90	−21.92	21.02
700	−0.90	−28.95	28.05

The insertion loss was calculated using $IL = P_{\text{in}} - P_{\text{out}}$. The insertion loss is small (0.5–1 dB) and nearly flat at low frequencies, rises slowly up to about 500 MHz, and then increases rapidly at higher frequencies (Figure 1). The identified filter type is therefore

$$\boxed{\text{Low-pass filter}}$$

The mean passband insertion loss (100–300 MHz) is 0.78 dB, so the 3-dB cutoff level is $0.78 + 3 = 3.78$ dB. Linear interpolation between 500 and 550 MHz gives

$$\boxed{f_c \approx 548 \text{ MHz}}$$

Taking a stop-band level of 20 dB, interpolation between 600 and 650 MHz gives

$$\boxed{f_s \approx 646 \text{ MHz}}$$

The passband insertion loss is therefore about 0.5–1 dB, and the filter roll-off from f_c to f_s occurs over roughly 100 MHz.

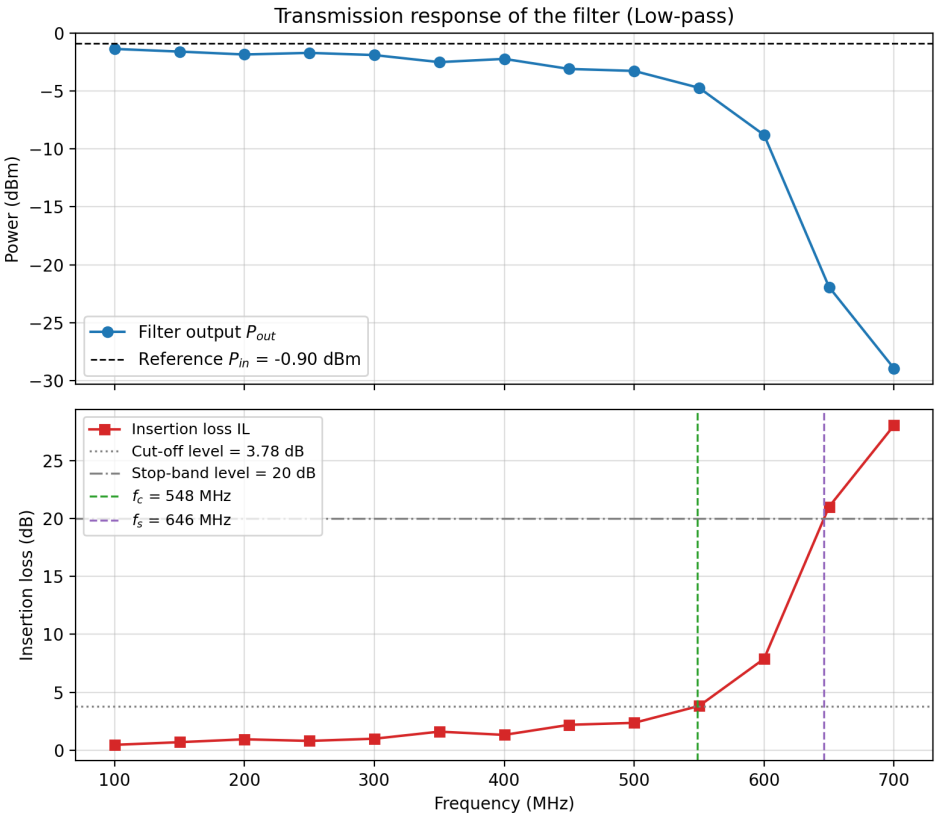


Figure 1: Filter transmission: output power (top) and insertion loss (bottom) versus frequency, with f_c and f_s marked.

6.2 Reflection Coefficient and Return Loss

Table 2: Filter reflection and return-loss measurements (flat passband region, 61–70 MHz).

Frequency (MHz)	$V_{\max}$ (mV)	$V_{\min}$ (mV)	V_{inc} (mV)	V_{ref} (mV)	RL (dB)
61–70	136	132	134	2	36.52

The reflection coefficient was obtained from

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{136 - 132}{136 + 132} = \frac{4}{268}.$$

The return loss was calculated using $RL = -20 \log_{10} |\Gamma|$.

Measured average reflection coefficient:

$$\boxed{|\Gamma| = 0.0149}$$

Measured return loss:

$$\boxed{RL = 36.5 \text{ dB}}$$

The corresponding VSWR is $(1 + 0.0149)/(1 - 0.0149) \approx 1.03$, indicating a very good match to 50 Ω in the passband.

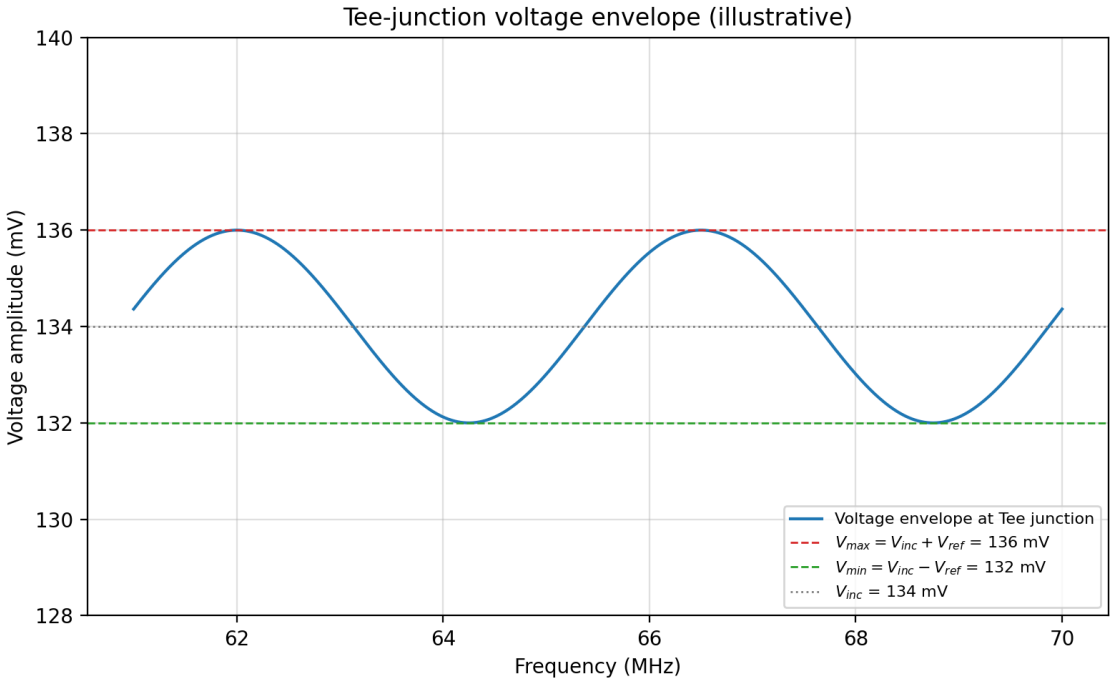


Figure 2: Left: illustrative Tee-junction voltage envelope (reconstructed from the measured $V_{\max}$ and $V_{\min}$ only). Right: return loss versus $|\Gamma|$ with the measured point.

6.3 Quarter-Wave Stub

The stub length was $L = 0.91$ m and the assumed velocity factor was $VF = 0.66$. The theoretical notch frequency is

$$f_{0,\text{calc}} = \frac{VF\,c}{4L} = \frac{0.66 \times 3 \times 10^8}{4 \times 0.91} = 54.40 \text{ MHz}.$$

Table 3: Quarter-wave stub measurements.

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	Attenuation (dB)
50	−0.90	−9.20	8.30
51	−0.90	−10.30	9.40
52	−0.90	−11.40	10.50
53	−0.90	−13.20	12.30
54	−0.90	−15.40	14.50
55	−0.90	−29.40	**28.50**
56	−0.90	−22.60	21.70
57	−0.90	−27.80	26.90
58	−0.90	−25.40	24.50
59	−0.90	−20.60	19.70
60	−0.90	−17.50	16.60
61	−0.90	−15.10	14.20

The calculated notch frequency was

$$\boxed{f_{0,\text{calc}} = 54.40 \text{ MHz}}$$

The experimentally measured notch frequency (maximum attenuation of 28.5 dB) was

$$\boxed{f_{0,\text{meas}} = 55 \text{ MHz}}$$

The difference is +0.60 MHz (+1.1%). A parabolic fit through the minimum and its two neighbours gives 55.2 MHz. The measured frequency corresponds to an effective velocity factor of $VF_{\text{eff}} = 4Lf_0/c = 0.667$, close to the assumed value of 0.66.

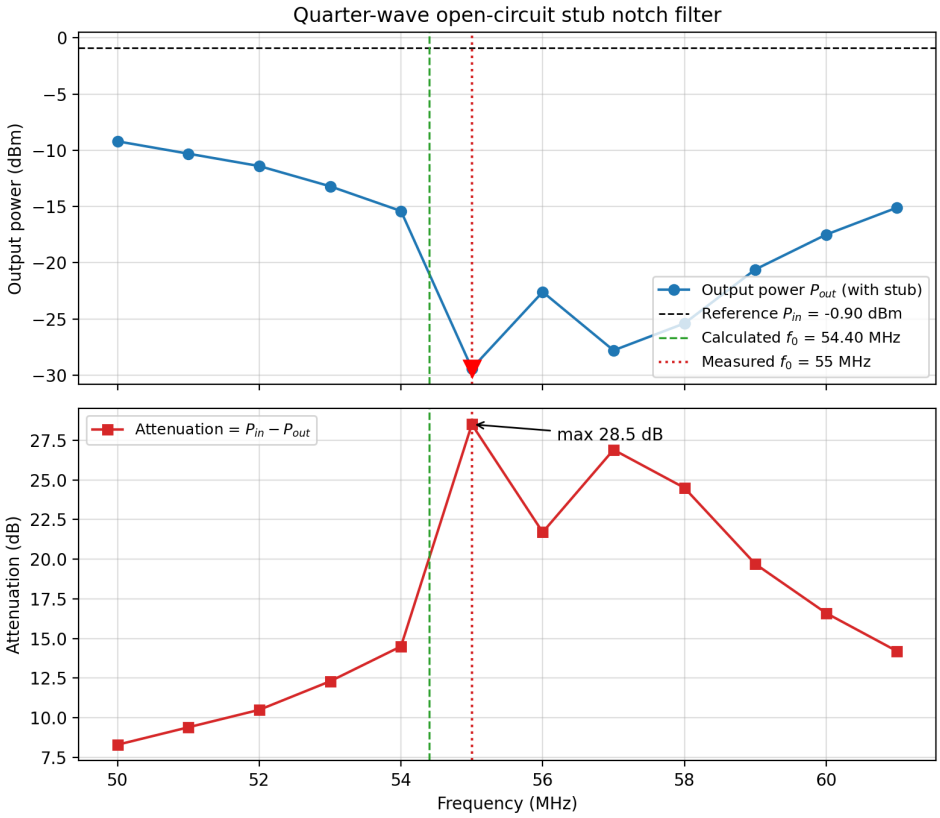


Figure 3: Quarter-wave stub: output power (top) and attenuation (bottom) versus frequency, with calculated and measured notch frequencies.

6.4 Mixer Characterization

The LO frequency was $f_{\text{LO}} = 10$ MHz with $P_{\text{LO}} = 1.04$ dBm, and the RF input power was constant at $P_{\text{RF}} = -0.90$ dBm.

Table 4: Mixer measurements.

f_{RF} (MHz)	f_{LO} (MHz)	Sideband	$f_{RF}+f_{LO}$ (MHz)	$\lvert f_{RF}-f_{LO}\rvert$ (MHz)	f_{IF} (meas.) (MHz)	P_{IF} (dBm)	L_{conv} (dB)
13	10	Up	23	3	23	−13.80	12.90
13	10	Down	23	3	3	−15.00	14.10
23	10	Up	33	13	33	−14.10	13.20
23	10	Down	33	13	13	−13.80	12.90
33	10	Up	43	23	43	−10.08	9.18
33	10	Down	43	23	23	−10.40	9.50
43	10	Up	53	33	53	−10.57	9.67
43	10	Down	53	33	33	−10.61	9.71

The conversion loss was calculated using $L_{conv} = P_{RF} - P_{IF}$. In every case the measured IF frequency equals either $f_{RF} + f_{LO}$ (up-conversion) or $|f_{RF} - f_{LO}|$ (down-conversion). The average conversion loss is 11.4 dB (11.2 dB for up-conversion and 11.6 dB for down-conversion), with a minimum of 9.18 dB ($f_{RF} = 33$ MHz, up) and a maximum of 14.10 dB ($f_{RF} = 13$ MHz, down).

$$\boxed{\overline{L_{conv}} \approx 11.4 \text{ dB}}$$

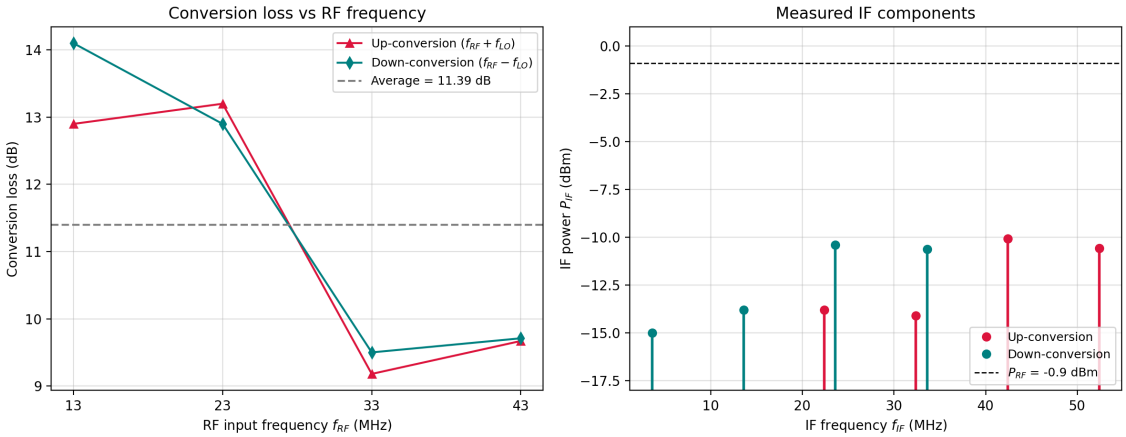


Figure 4: Left: conversion loss versus RF frequency. Right: measured IF components.

6.5 Power Splitter

Table 5: Two-way power splitter measurements.

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	L_1 (dB)	L_2 (dB)
33	−0.90	−4.42	−4.40	3.52	3.50
43	−0.90	−4.46	−4.44	3.56	3.54
53	−0.90	−4.47	−4.43	3.57	3.53

For an ideal two-way splitter, each output is expected to be approximately 3 dB below the input. Average measured losses:

$$\boxed{L_1 = 3.55 \text{ dB}}$$

$$\boxed{L_2 = 3.52 \text{ dB}}$$

The two outputs agree to within 0.04 dB (good balance), and each has about 0.5 dB excess loss over the ideal 3 dB, which is attributed to the insertion loss of the device, connectors and cables. The loss is nearly constant from 33 to 53 MHz.

6.6 Power Splitter

The two-way power splitter was characterized by measuring the input power and the power at each of the two output ports at three different frequencies. The measured values and calculated path losses are shown in Table 6.

Table 6: Two-way power splitter measurements and calculated path losses.

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	L_1 (dB)	L_2 (dB)
33	-0.90	-4.42	-4.40	3.52	3.50
43	-0.90	-4.46	-4.44	3.56	3.54
53	-0.90	-4.47	-4.43	3.57	3.53

The path loss for each output port was calculated using

$$L_1 = P_{\text{in}} - P_{\text{out1}}, \qquad L_2 = P_{\text{in}} - P_{\text{out2}}.$$

For an ideal two-way power splitter, each output is expected to be approximately 3 dB below the input power. The average measured losses are

$$\boxed{L_1 = 3.55 \text{ dB}}$$

and

$$\boxed{L_2 = 3.52 \text{ dB}}.$$

The difference between the two output losses is only about 0.03 dB, indicating that the two output ports are closely balanced. The measured loss is approximately 0.5 dB higher than the ideal 3 dB splitting loss. This additional loss can be attributed to the insertion loss of the splitter and the losses associated with the connectors and cables.

The measured path losses remain nearly constant over the frequency range from 33 to 53 MHz. The variation of the two output path losses with frequency is shown in Fig. 5.

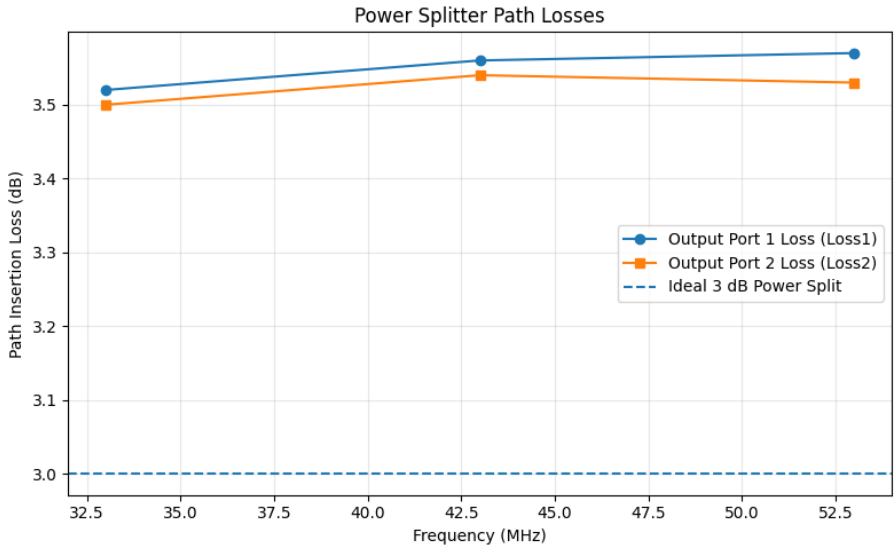


Figure 5: Measured path loss of the two output ports of the two-way power splitter as a function of frequency.

Power Combiner

The power combiner was characterized by applying two signals to its input ports and measuring the combined output. The two input signals were set to

$$f_1 = f_2 = 10 \text{ MHz},$$

with powers

$$P_1 = -0.90 \text{ dBm}, \qquad P_2 = 1.64 \text{ dBm}.$$

The corresponding peak voltages were measured as

$$V_1 = 201.8 \text{ mV}, \qquad V_2 = 270.4 \text{ mV}.$$

For coherent, in-phase signals, the voltages add according to

$$V_{\text{sum}} = V_1 + V_2.$$

Therefore,

$$V_{\text{sum}} = 201.8 + 270.4 = 472.2 \text{ mV}.$$

For a 50 Ω system, the corresponding ideal combined power is

$$P_{\text{ideal}} = \frac{V_{\text{sum}}^2}{2R} = \frac{(0.4722)^2}{2(50)} = 2.23 \text{ mW}.$$

Converting this power to dBm gives

$$\boxed{P_{\text{ideal}} = 3.48 \text{ dBm}}.$$

The measured combined output power was

$$\boxed{P_{\text{meas}} = 3.05 \text{ dBm}}.$$

Thus, the measured output was approximately 0.43 dB lower than the ideal in-phase vector sum. This difference can arise from the relative phase between the two input signals and from losses in the combiner and measurement setup.

The measured result is closer to the coherent voltage-addition case than to an incoherent power addition, demonstrating the vector nature of RF signal combining. The corresponding relative phase between the two input signals is approximately 36°.

The measured and calculated power-combiner results are shown in Fig. 6.

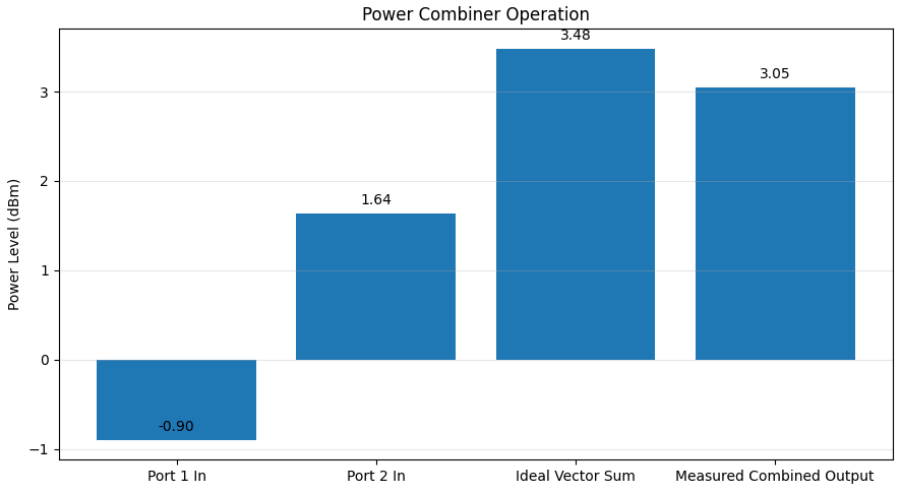


Figure 6: Power-combiner output showing the measured combined power compared with the calculated ideal coherent-combination result.

6.7 Amplifier Gain

The amplifier gain was measured over the frequency range from 100 to 200 MHz with a constant input power of −20 dBm. The gain was calculated using

$$G = P_{\text{out}} - P_{\text{in}}.$$

Table 7: Amplifier gain versus frequency ($P_{\text{in}} = -20$ dBm).

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
100	−20	2.68	22.68
110	−20	2.65	22.65
120	−20	2.58	22.58
130	−20	2.52	22.52
140	−20	2.43	22.43
150	−20	2.34	22.34
160	−20	2.30	22.30
170	−20	2.27	22.27
180	−20	2.24	22.24
190	−20	2.23	22.23
200	−20	2.22	22.22

The measured gain decreases gradually from 22.68 dB at 100 MHz to 22.22 dB at 200 MHz. The total variation is approximately 0.46 dB, indicating that the amplifier gain remains relatively stable over the measured frequency range.

The measured gain variation with frequency is shown in Fig. 7.

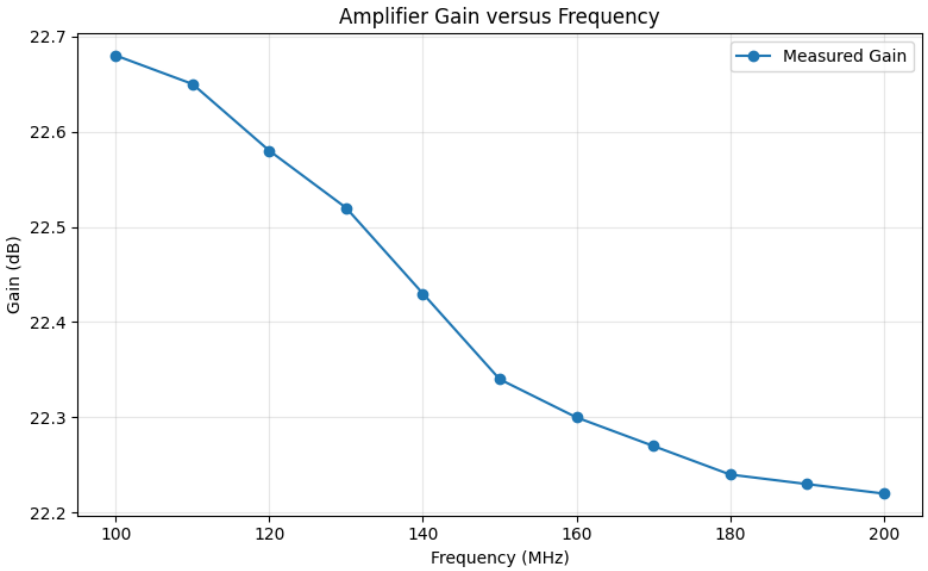


Figure 7: Measured amplifier gain as a function of frequency for $P_{\text{in}} = -20$ dBm.

6.8 1-dB Compression Point

The amplifier compression characteristics were measured by increasing the input power and recording the corresponding output power. The calculated gain is also shown in Table 8.

Table 8: Amplifier compression measurements.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)	P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
−20	2.68	22.68	−6	16.23	22.23
−19	3.67	22.67	−5.5	16.28	21.78
−18	4.64	22.64	−5.4	16.32	21.72
−17	5.64	22.64	−5.3	16.47	21.77
−16	6.62	22.62	−5	16.70	21.70
−15	7.62	22.62	−4	17.20	21.20
−14	8.60	22.60	−3	17.40	20.40
−13	9.62	22.62	−2	17.54	19.54
−12	10.58	22.58	−1	17.61	18.61
−11	11.54	22.54	0	16.28	16.28
−10	12.52	22.52			
−9	13.53	22.53			
−8	14.42	22.42			
−7	15.20	22.20			

The small-signal gain, averaged over the five lowest input powers from -20 to -16 dBm, is approximately 22.65 dB. Therefore, the gain corresponding to the 1-dB compression point is

$$G_{1\mathrm{dB}} = 22.65 - 1 = 21.65 \text{ dB}.$$

The gain is 21.70 dB at $P_{\mathrm{in}} = -5$ dBm and 21.20 dB at $P_{\mathrm{in}} = -4$ dBm. Linear interpolation gives

$$P_{1\mathrm{dB,in}} \approx -4.9 \text{ dBm}.$$

The corresponding output power is approximately

$$P_{1\mathrm{dB,out}} \approx 16.8 \text{ dBm}.$$

Thus,

$$\boxed{P_{1\mathrm{dB,in}} \approx -4.9 \text{ dBm}}$$

and

$$\boxed{P_{1\mathrm{dB,out}} \approx 16.8 \text{ dBm}}.$$

The amplifier output reaches a maximum of approximately 17.6 dBm in the measured range. The gain compression and output-power characteristics are shown together in Fig. 8.

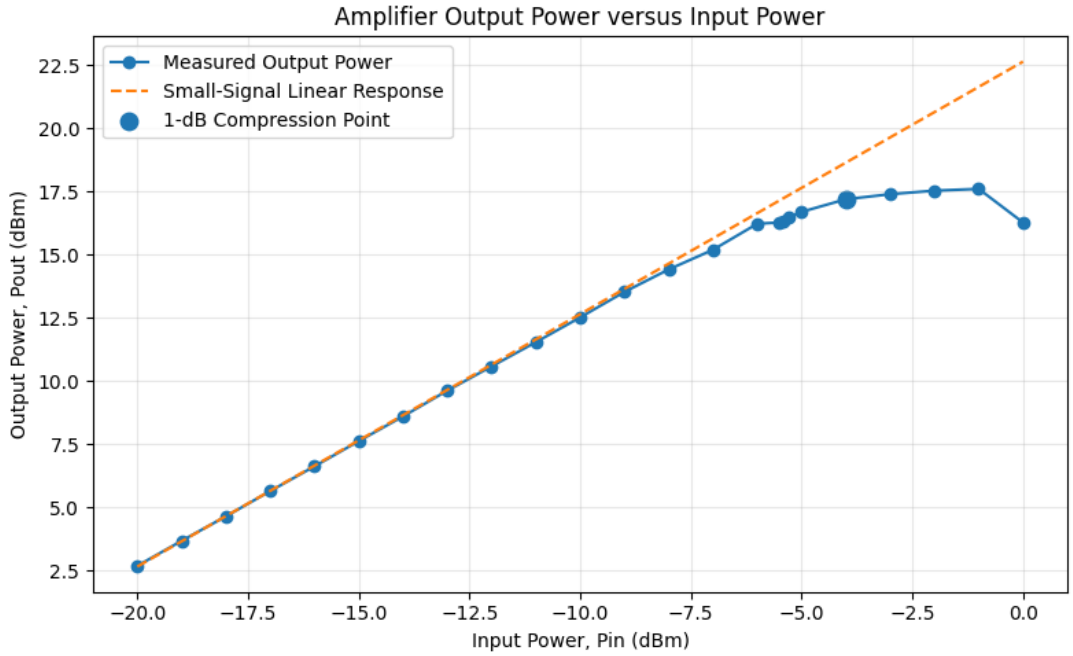


Figure 8: Amplifier gain and output power as functions of input power. The two characteristics are shown in the same figure.

No harmonic power measurements were included in the data set used for this analysis.

6.9 Third-Order Intercept Point

A two-tone test was performed using input frequencies

$$f_1 = 10 \text{ MHz}, \qquad f_2 = 13 \text{ MHz}.$$

The third-order intermodulation products occur at

$$2f_1 - f_2 = 7 \text{ MHz}$$

and

$$2f_2 - f_1 = 16 \text{ MHz}.$$

The measured fundamental and third-order intermodulation powers are listed in Table 9.

Table 9: Third-order intermodulation measurements ($f_1 = 10$ MHz, $f_2 = 13$ MHz).

Input Power per Tone (dBm)	Fundamental Output (dBm)	Third-Order Product (dBm)
−20	0.74	−61.20
−18	2.76	−57.60
−16	4.70	−51.20
−14	6.67	−45.09
−12	8.62	−38.25
−10	10.47	−25.07

The fundamental output and third-order intermodulation product were plotted against the input power. For an ideal two-tone amplifier response, the fundamental component increases with a slope of 1, whereas the third-order intermodulation product increases with a slope of 3.

Using linear fits with the slopes fixed at 1:1 and 3:1, the average intercepts were obtained as

$$C_1 = 20.66 \text{ dB}$$

for the fundamental component and

$$C_3 = -1.40 \text{ dB}$$

for the third-order intermodulation component.

The input-referred TOI is therefore

$$P_{\text{in,TOI}} = \frac{C_3 - C_1}{1 - 3} = 11.03 \text{ dBm}.$$

The corresponding output-referred TOI is

$$P_{\text{out,TOI}} = P_{\text{in,TOI}} + C_1 = 31.69 \text{ dBm}.$$

Hence,

$$\boxed{P_{\text{in,TOI}} \approx 11.0 \text{ dBm per tone}}$$

and

$$\boxed{TOI_{\text{out}} \approx 31.7 \text{ dBm}}.$$

The measured TOI characteristic is shown in Fig. 9.

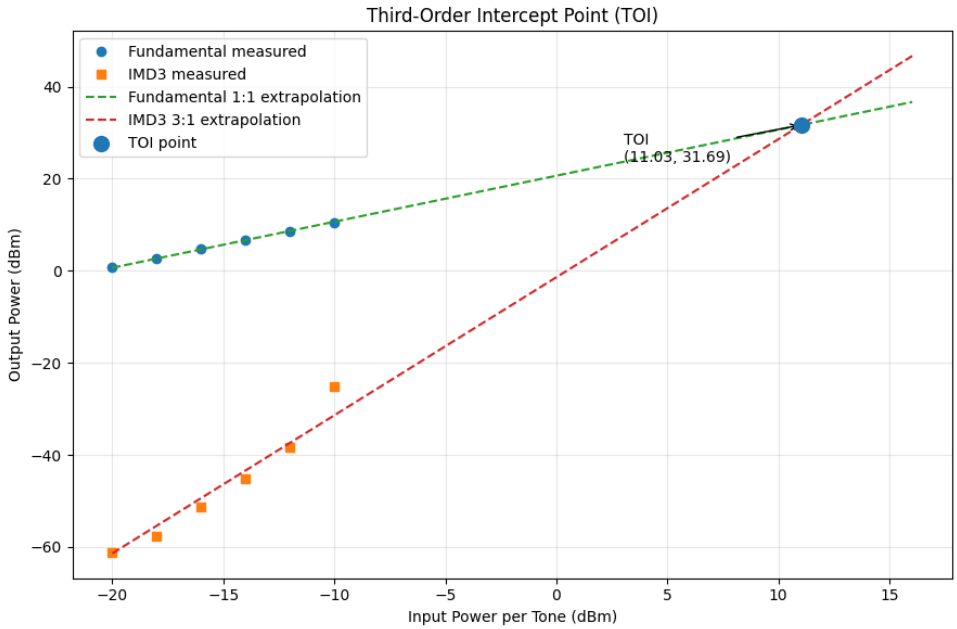


Figure 9: Third-order intercept point obtained from the fundamental and third-order intermodulation output powers.

The last measured IM3 point at $P_{\text{in}} = -10$ dBm lies considerably above the 3:1 trend. If this point is excluded from the fit, the extrapolated output TOI is approximately 32.5 dBm. Therefore, the TOI extracted from the available data has an uncertainty of roughly ± 1 dB.

6.10 Summary of Principal Measurements

Table 10: Summary of results.

Quantity	Symbol	Measured/calculated value
Low-pass cutoff and stop-band frequencies	f_c, f_s	548 MHz, 646 MHz
Band-pass/notch frequencies of filter	f_1, f_0, f_2	Not applicable (low-pass filter)
Filter insertion loss	IL	0.46–0.99 dB up to 300 MHz; 3.82 dB at 550 MHz; 28.05 dB at 700 MHz
Filter reflection coefficient	$\lvert\Gamma\rvert$	0.0149
Filter return loss	RL	36.5 dB (VSWR ≈ 1.03)
Stub notch frequency	f_0	Calculated 54.40 MHz; measured 55 MHz
Mixer frequencies	$f_{\mathrm{RF}}, f_{\mathrm{LO}}, f_{\mathrm{IF}}$	$f_{\mathrm{LO}} = 10$ MHz; $f_{\mathrm{RF}} = 13, 23, 33, 43$ MHz; $f_{\mathrm{IF}} = f_{\mathrm{RF}} \pm 10$ MHz; mean $L_{\mathrm{conv}} = 11.4$ dB
Power-splitter output powers	$P_{\mathrm{out1}}, P_{\mathrm{out2}}$	$-4.42/-4.40$, $-4.46/-4.44$, $-4.47/-4.43$ dBm at 33, 43, 53 MHz
Amplifier input, output and gain	$P_{\mathrm{in}}, P_{\mathrm{out}}$, Gain	-20 dBm; 2.68 to 2.22 dBm; 22.68 to 22.22 dB (100–200 MHz)
1-dB compression point	P_{1dB}	-4.9 dBm in, 16.8 dBm out
Third-order intercept point	TOI	31.7 dBm (output), 11.0 dBm (input per tone)

7 Conclusion

The RF filter, mixer, power splitter/combiner and amplifier were experimentally characterized.

- The filter is a **low-pass** filter with $f_c \approx 548$ MHz, passband insertion loss of about 0.5–1 dB and a return loss of 36.5 dB.
- The quarter-wave stub produced a notch at 55 MHz (28.5 dB attenuation) against 54.4 MHz calculated.
- The mixer produced the expected sum and difference frequencies with an average conversion loss of 11.4 dB.
- The two-way splitter gave about 3.5 dB loss on each arm, close to the ideal 3 dB, and the combiner output was within 0.43 dB of the ideal vector sum.
- The amplifier gain was about 22.7 dB (100–200 MHz), with $P_{\mathrm{1dB,in}} \approx -4.9$ dBm ($P_{\mathrm{1dB,out}} \approx 16.8$ dBm) and $TOI_{\mathrm{out}} \approx 31.7$ dBm.

- Harmonic levels: [**insert measured** $2f$**,** $3f$ **powers**]; these grow faster than the fundamental as the amplifier is driven towards compression.
- Main uncertainty sources: cable and connector loss and mismatch, spectrum-analyser and generator power accuracy, oscilloscope loading and reading resolution in the Tee method, the 1 MHz to 50 MHz frequency steps, and (for the TOI) the noise floor and compression affecting the IM3 points.